Lower Nicola Watershed Connectivity Restoration Plan: 2021 - 2040

Canadian Wildlife Federation

11-02-2026

Table of contents

Project Overview

Purpose

The purpose of the Watershed Connectivity Restoration Plan (WCRP) is to systematically identify and assess potential barriers to salmon migration in the Lower Nicola Watershed. By conducting both desktop and field assessments, the project aims to evaluate the extent of migration barriers caused by manmade structures and identify opportunities for increasing connectivity for salmonid species. This assessment will guide the prioritization of barriers for remediation, ensuring that the most impactful sites are addressed first to enhance salmon habitat and improve migration conditions.

Current Connectivity Status

In the Lower Nicola watershed, km of Pacific Salmon habitat are currently connected to the ocean and #km is disconnected from it. This represents that % of the # km of total habitat is connected.

In the Lower Nicola Watershed, # structures potentially disconnect Pacific Salmon habitat. Of these, # are identified as barriers in need of rehabilitation (priority barriers), # barriers that do not warrant rehabilitation (non-actionable), and # require further field assessment.

Habitat Accumulation Curve

Map

Progress Summary

The planning process for the Lower Nicola River Watershed Connectivity Rehabilitation Plan began in 2020. Since that time, 244 structures have been assessed to determine their fish passage status, and 19 habitat confirmations have been completed. In total, 4 barriers have been rehabilitated, resulting in improved or restored access to 12.9 km of spawning habitat and 17.4 km of rearing habitat for focal species. In addition, spawning and rearing habitat was

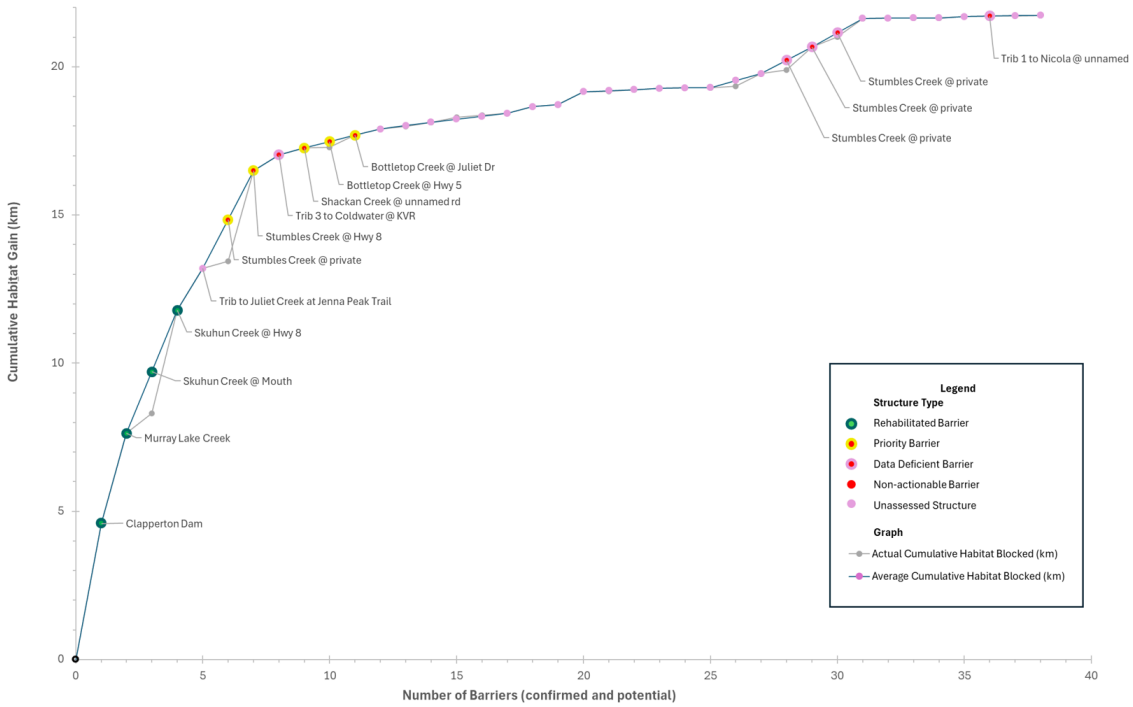

Figure 1: Figure 1: Habitat Accumulation Curve (HAC) showing structures in the Lower Nicola watershed as of x date. Structures are ranked based on how much (km) habitat is upstream to the next structure or set of structures and that would be reconnected if the structure(s) became passable. Sets of structures are indicated by a grey line below the curve, indicating that gains from addressing the downstream structure alone would be less than the combined gains from addressing all structures in the set. Habitat upstream of data-deficient structures is considered disconnected until field assessments are completed. Structures that were excluded as passable are not shown; however, rehabilitated barriers are shown to demonstrate the connectivity gains achieved through implementation of this plan.

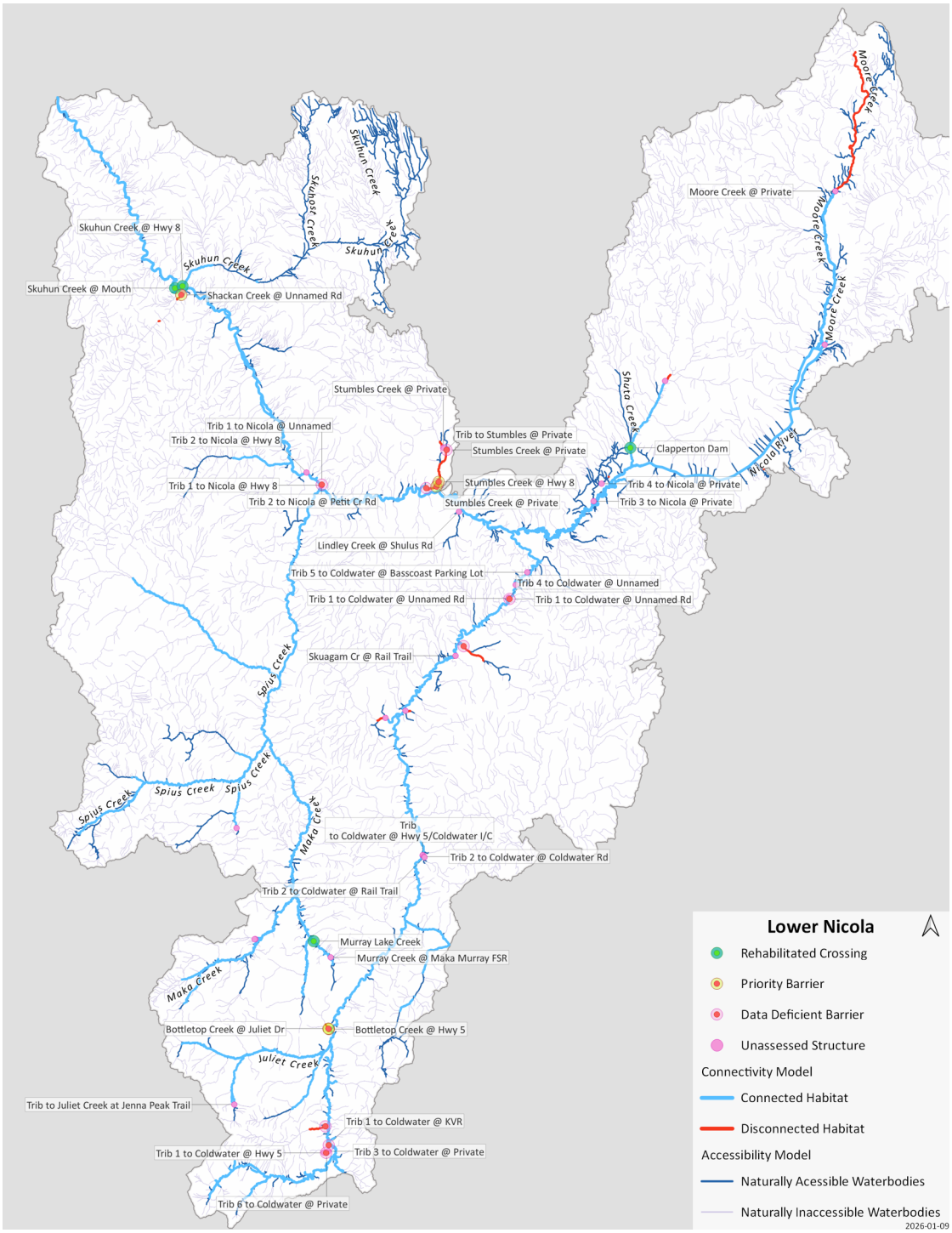

Figure 2: Map of focal species habitat and structures that are confirmed or potential barriers to fish passage in the Lower Nicola watershed as of x date. Structure data were obtained from the Canadian Aquatic Barriers Database (aquaticbarriers.ca). The accessibility model represents areas of the watershed that focal species could access naturally in the absence of anthropogenic barriers. The habitat model represents the subset of accessible areas that may be used by focal species for key habitat function(s). Available local knowledge and data were incorporated and overrule habitat model results. Thick light-blue lines represent habitat that is currently connected, whereas thick red lines represent habitat considered to be disconnected because it is upstream of confirmed barriers or data-deficient structures. The y-axis represents the amount of habitat (km) blocked by each structure, discounted by 50% for first-order streams, and by passability (25%, 50%, 0r 75%) of partial barriers. Barriers that were rehabilitated through implementation of this plan are shown, but

mapped and field-verified in tributaries with known or suspected key habitat values for focal species. Three drone flights on Guichon Creek, Stumbles, Creek, and the Nicola mainstem upstream of the Coldwater confluence, and one helicopter flight of the upper Coldwater River have been completed to-date by the Lower Nicola WCRP working group, with additional ongoing lateral and thermal connectivity work having been completed by other entities in the watershed by means of drone flights, helicopter flights, and field surveys completed by ground crews.

1 Vision and Scope

Purpose

The following Watershed Connectivity Restoration Plan (WCRP) represents the culmination of a one-year collaborative planning effort, including action implementation, for the Lower Nicola River watershed, the overall aim of which is to build collaborative partnerships within the watershed to reduce the threat of aquatic barriers to migratory fish and the livelihoods that they support. This 10-year plan was developed to identify priority strategies that the Lower Nicola River WCRP planning team (see Table **??**) proposes to undertake between 2021-2031 to conserve and restore fish passage in the watershed, through lateral and thermal barrier rehabilitation, crossing rehabilitation, and barrier prevention strategies.

WCRPs are long-term, actionable plans that blend local stakeholder and rightsholder knowledge with innovative geographical information systems (GIS) analyses to gain a shared understanding of where restoration efforts will have the greatest benefit for migratory fish. The planning process is inspired by the Conservation Standards (v.4.0), which is a conservation planning framework that allows planning teams to systematically identify, implement, and monitor strategies to apply the most effective solutions to high priority conservation problems. There is a rich history of fish and fish habitat conservation and restoration work in the Lower Nicola River watershed that this WCRP builds upon and aims to complement over the length of the plan. This includes work undertaken by the Scw'exmx Tribal Council and the four member or affiliate nations (Coldwater, Upper Nicola, Shackan and Nooaitch bands), the Lower Nicola Indian Band, the Nicola Watershed Governance Project including the Salmon Ecosystem Table and Nicola Restoration Committee, the Nicola Basin Collaborative, and the Risk Assessment Methodology for Salmon (RAMS). The planning team will aim to work with the Nicola Watershed Governance Project and the Nicola Basin Collaborative to promote coordination, decision-making, and implementation related to this plan.

The planning team compiled existing structure location and assessment data, habitat data, and previously identified priorities in the watershed, and combined this with local and Indigenous knowledge to create a strategic watershed-scale plan to improve freshwater connectivity. The Lower Nicola River WCRP planning team applied the WCRP planning framework to define the "thematic" scope of freshwater connectivity and refine the "geographic" scope to identify the portions of the watershed where connectivity restoration efforts will take place. Additionally, the team selected focal species, assessed the current key habitat connectivity status of the

watershed, defined concrete goals for gains in connectivity, sundertook an iterative structure-ranking process to identify a list of priority barriers for rehabilitation to achieve those goals (see Table **??**). Although the current version of this plan is based on the best-available information at the time of publishing, WCRPs are intended to be "living plans" that are updated regularly as new information becomes available, or if local priorities and contexts change. As such, this document should be interpreted as a current "snapshot" in time, and future iterations of this WCRP will build upon the results presented in this plan to continuously improve the practice of aquatic barrier rehabilitation for migratory fishes in the Lower Nicola River watershed. For more information on how WCRPs are developed, see Mazany-Wright et al. (2021b).

Vision

Healthy, well-connected stream networks within the Lower Nicola River watershed support thriving populations of migratory fish. In turn, these fish provide the continued sustenance, cultural, and ceremonial needs of the Nlaka'pamux/Scw'exmx and Syilx peoples, as they have since time immemorial. Both residents and visitors to the watershed work together to mitigate the negative effects of aquatic barriers, improving the resiliency of the stream network for the benefit and appreciation of all.

Scope

The primary geographic scope of this WCRP is the Lower Nicola River watershed, located in the Thompson drainage basin of the Fraser River system in south-central British Columbia drainage area 376,064 ha; (Figure **??**). The scope constitutes the Lower Nicola "watershed group" as defined by the British Columbia Freshwater Atlas (FWA), which excludes the Guichon Creek drainage and the Nicola River and Quilchena Creek drainages upstream of Nicola Lake. A consistent spatial framework was necessary to undertake a watershed-selection process at the provincial scale to identify target watersheds to improve freshwater connectivity for salmonids. The Lower Nicola River watershed was identified by the B.C. Fish Passage Restoration Initiative as one of four target watersheds for WCRP development (Mazany-Wright et al. (2021c)). Culturally and economically important populations of Chinook Salmon (*Oncorhynchus tshawtyscha*), Coho Salmon (*Oncorhynchus kisutch*), and the anadromoius life history of Rainbow Trout commonly referred as steelhead (*Oncorhynchus mykiss*) are all found in the watershed, which historically supported Indigenous sustenance and trading economies Table 1.1 (**?@tbl-targspec**; Lower Nicola Indian Band (2015), ESSA Technologies Ltd. and Fraser Basin Council (2019), Coldwater Band (2021)).

Nłeʔkepmxcín (Nlaka'pamuxcin) Name	nqilxʷcn (nsyilxcən) Name	Common Name	Scientific N
k'ʷy'íʔe/pəqéłus	ntytyix	Chinook Salmon	Oncorhyncl

Nłeʔkepmxcín (Nlaka'pamuxcin) Name	nqilxʷcn (nsyilxcən) Name	Common Name	Scientific N
sx̓ȧ́yqs	kisúʔ	Coho Salmon	Oncorhyncl
cóʕʷłeʔ	wəyqwəyʕaća̓ʔ	Steelhead	Oncorhyncl

The Lower Nicola River watershed comprises parts of the traditional territory of the Nlaka'pamux/Scw'exmx and Syilx Peoples, represented by the Scw'exmx Tribal Council, the four member nations (Coldwater Band, Nooaitch Band, Shackan Band, and Upper Nicola Band), and the individual nations of the Lower Nicola Indian Band and the Cook's Ferry Band. The Nlaka'pamux/Scw'exmx and Syilx peoples steward the land and the waters of the Lower Nicola River watershed. The planning team will pursue early, meaningful, and continued engagement with First Nations communities involved in work and projects related to this plan. It will be necessary to receive permission from the communities for any work to occur on their territory.

The primary geographic scope of this WCRP was further refined by identifying "naturally accessible" waterbodies, which are defined as waterbodies that focal species should be able to access in the absence of human-made barriers (Figure **??**). Naturally accessible waterbodies were spatially delineated using fish species observation and distribution data, as well as data on "natural barriers", which are waterfalls greater than 5 m in height, gradient barriers based on literature-derived; species-specific swimming abilities. "Watershed exclusion areas" were also identified, which are portions of the watershed where barrier rehabilitation efforts should not occur due to additional considerations. These maps were explored by the planning team to incorporate additional local knowledge, ensure accuracy, and finalize the constraints on naturally accessible waterbodies. The planning team identified a few tributaries to the mainstem Nicola River as watershed exclusion areas due to intermittent or insufficient flows to support restoring connectivity for the focal species, including Hamilton Creek and agricultural irrigation ditches just downstream of Nicola Lake Dam. Additionally, Stumplake Creek and Peter Hope Creek were identified as watershed exclusion areas due to the presence of invasive Yellow Perch (*Perca flavescens*). It is unclear whether existing structures located in these systems will be effective in preventing the downstream spread of Yellow Perch, but the planning team advised maintaining the structures for the time being. All waterbodies not identified as naturally accessible were removed from the project scope. The resulting constrained geographic scope formed the foundation for all subsequent analyses and planning steps, including mapping and modelling key habitat, quantifying the current connectivity status, goal setting, and action planning (Mazany-Wright, N and Norris, S. M. and Lapointe, N. W. R. and Rebellato, B. (2021a)).

The thematic scope of this WCRP is freshwater connectivity. Connectivity influences critical components of freshwater ecosystem structure and function, including aquatic species dispersal and migration, the transport of energy and matter (e.g., nutrient cycling and sediment

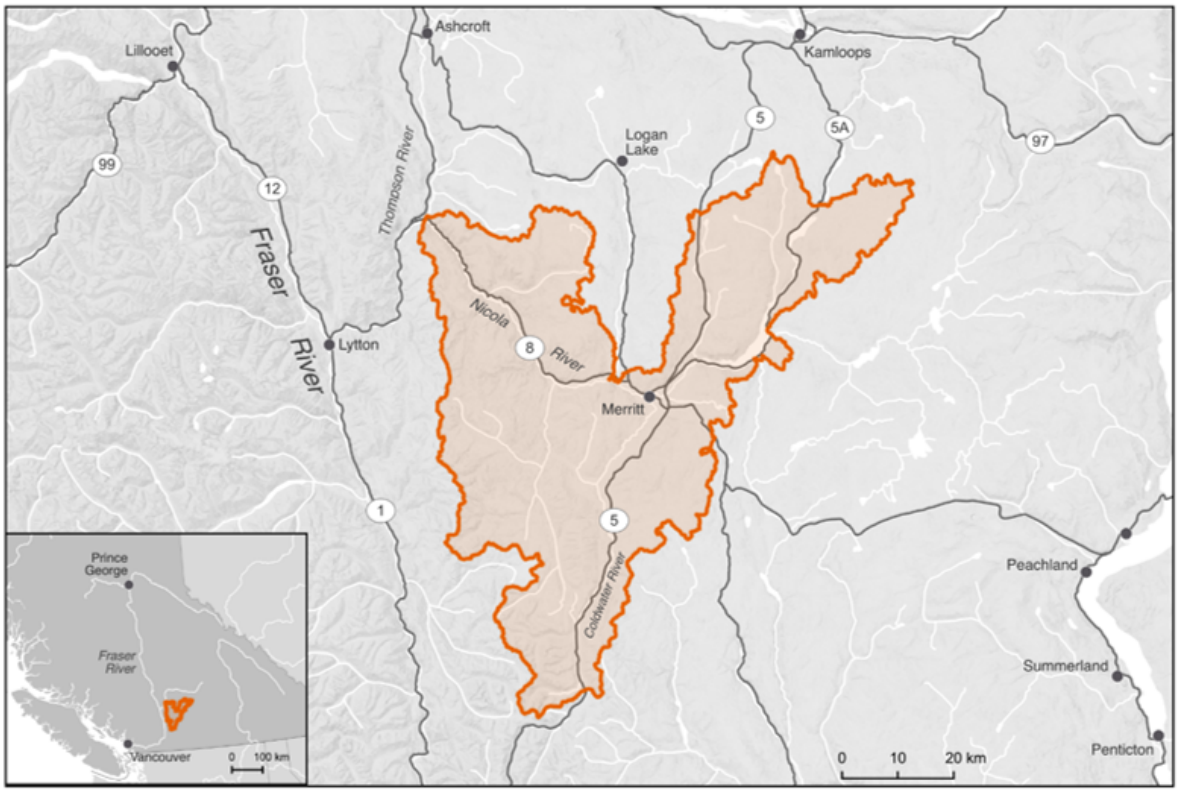

Figure 1.1: The primary geographic scope - the Lower Nicola River watershed, excluding the Guichon Creek drainage and the Nicola River and Quilchena Creek drainages upstream of Nicola Lake

Table 1.1: **?(caption)**

Table 1.1: Focal species in the Lower Nicola River watershed. The Nłe kepmxcín (Nlaka'pamuxcin), nqilx cn (nsyilxcən), and Western common and scientific species names are provided.

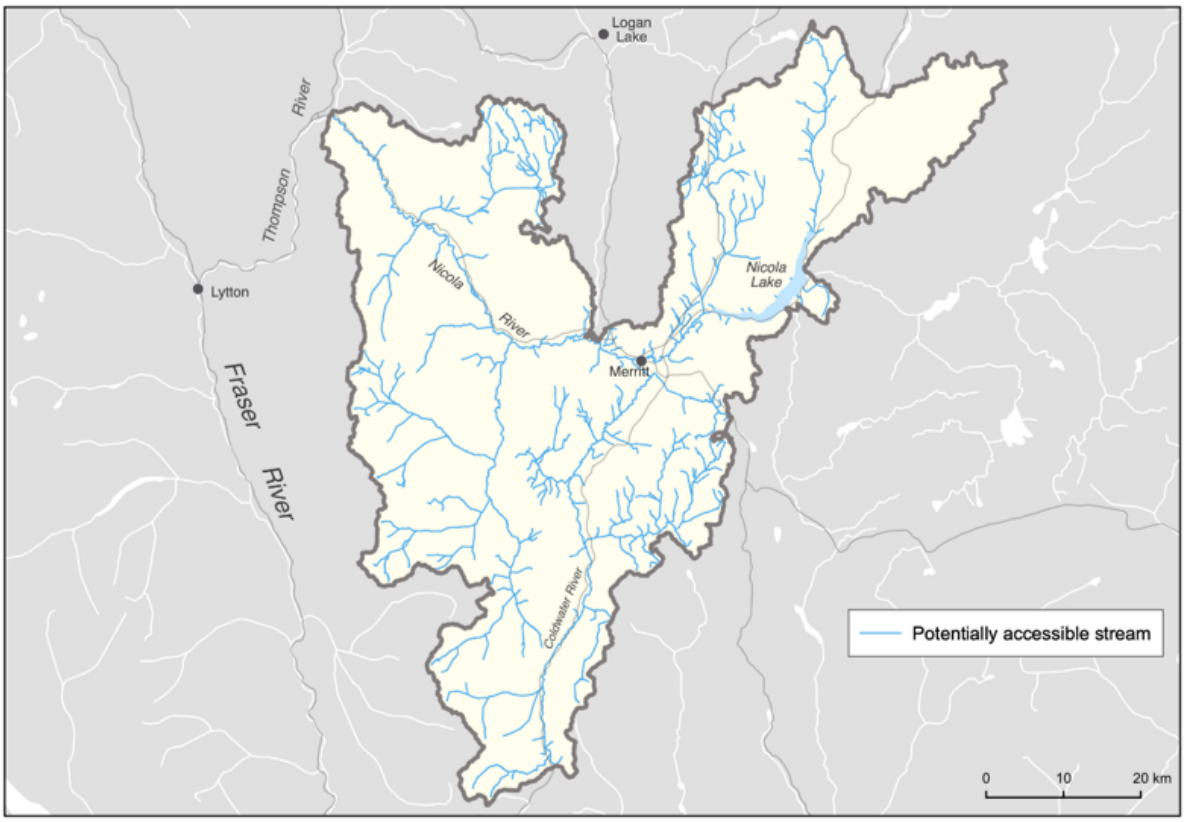

Figure 1.2: Naturally accessible waterbodies within the Lower Nicola River watershed. These do not represent key habitat, but rather identify the waterbodies within which habitat modelling and barrier mapping and prioritization were undertaken.

flows), and temperature regulation (Seliger and Zeiringer (2018)). Though each of these factors are important when considering the health of a watershed, for the purposes of this WCRP the term "freshwater connectivity" is defined as the degree to which focal species can disperse or migrate freely through freshwater systems. Connectivity can be disrupted by physical structures in the longitudinal (i.e., upstream-downstream) and lateral (i.e., connectivity between the mainstem and adjacent wetlands, floodplains, side channels, and off-channel habitat) planes, including dams, weirs, stream crossings, dykes, linear infrastructure, and debris flows. Freshwater systems can also be disconnected by "physiological" barriers that prevent the free dispersal of species, including thermal (i.e., reaches where stream temperatures are too high) or flow (i.e., reaches where stream flow is insufficient to support the requirements of any life stage) barriers.

The broader Nicola basin is a temperature- and flow-sensitive watershed, and both factors significantly affect connectivity for fish species in the Lower Nicola (ESSA Technologies Ltd. and Fraser Basin Council (2019)). The changing thermal regime of the Lower Nicola River watershed is a growing concern, with two scales of thermal impacts occurring within the watershed — watershed-scale changes in thermal regimes and localized barriers preventing access to thermal refugia.

The watershed-scale changes to the thermal regime are linked to several landscape-scale drivers including increases in water withdrawals, changes in land use, deforestation (due to resource extraction and mountain pine beetle infestations), and climate change, which are exacerbated by subsequent changes to channel-forming processes (ESSA Technologies Ltd. and Fraser Basin Council (2019)). These changes have created resulted in major wildfires and flooding occurring in 2017, 2020 and 2021, which have in turn accelerated changes to channel-forming processes, flows, and stream temperatures. Thermal barriers that prevent access to headwater reaches from the mainstem channels form annually along the valley floor. This has resulted in the adaptation of an early-migrant Chinook Salmon population, which uses the upper portions of Spius Creek and tributaries and the upper reaches of the Coldwater River, where access to the spawning grounds requires passing through the lower sections before the stream temperatures create physiological thermal barriers. These returning adults pass through the lower reaches in May and June as the freshet starts to abate, then hold in deeper pools for two months until they spawn in mid-to-late August (R. Bailey, Nooaitch Band, pers. comm.).

Localized thermal disconnectivity in the lateral dimension occurs when rearing and out-migrating juveniles are unable to access side-channel and off-channel thermal refugia and holding pools due to changing channel processes exacerbated by upland management and the development of linear infrastructure including dykes, roads, railways, and trails. These groundwater-serviced habitats provide thermal refuge for juvenile fish in the watershed; and can also provide critical refuge for returning spawners. When these lateral refugia become disconnected from mainstem channels, lethal stream temperatures can cause juvenile die-offs. Activities that restore and protect connectivity to these lateral refugia, particularly those cooled by groundwater sources, can help mitigate thermal disconnectivity in the watershed.

Mitigation of the landscape-scale drivers of changes to thermal regimes are outside of the scope of this plan. Although vital for the long-term resilience of the watershed, the broad land-use patterns, over-subscription of water withdrawals, and climate change that are causing chronic thermal issues in the watershed require more complex, coordinated, and resource-intensive solutions than this planning process can provide. This plan is intended to focus on the direct rehabilitation and prevention of localized, physical barriers to lateral thermal refugia and longitudinal connectivity to maintain fish passage to spawning, rearing, and refuge habitat. Lateral– and associated thermal– connectivity was identified by the planning team as the primary connectivity concern in the watershed within the defined scope of this plan.

Focal Species

Focal species represent the ecologically and culturally important species for which habitat connectivity is being conserved or restored in the watershed. In the Lower Nicola River watershed, the planning team selected anadromous salmonids as the focal guild, which comprises Chinook Salmon, Coho Salmon, and steelhead. The selection of these focal species was driven primarily by the focal species of the primary funds supporting this planning work. The planning team also identified other culturally and ecologically important species within the watershed to consider for inclusion in future iterations of the WCRP, including kokanee (*Oncorhynchus nerka*), Bull Trout (*Salvelinus confluentus*), Rainbow Trout (*Oncorhynchus mykiss*), Mountain Whitefish (*Prosopium williamsoni*), Lake Whitefish (*Coregonus clupeaformis*), Burbot (*Lota lota*), and Pink Salmon (*Oncorhynchus gorbuscha*).

Anadromous Salmonids

Anadromous salmonids are cultural and ecological keystone species that contribute to productive ecosystems by contributing marine-derived nutrients to the watershed and forming an important food source for bears and other species (Schindler and Quay. (2003)). Salmon and steelhead have enduring food, social, and ceremonial value for First Nations in the Lower Nicola River watershed – having sustained life, trading economies, and culture for the Nlaka'pamux/Scw'exmx and Syilx peoples since time immemorial (Lower Nicola Indian Band (2015), ESSA Technologies Ltd. and Fraser Basin Council (2019), Coldwater Band (2021)). The harvest and processing of these species have helped pass knowledge and ceremony to future generations (Fraser Basin Council (n.d.), Lower Nicola Indian Band (2015)).

Anadromous salmonid populations in the Lower Nicola River watershed have declined significantly since the mid-1980s, leading First Nation communities to voluntarily reduce their harvest (ESSA Technologies Ltd. and Fraser Basin Council (2019)). The Nlaka'pamux/Scw'exmx and Syilx peoples have always been stewards of the lands, resources, and fisheries in their traditional territories through an interconnected relationship based on respect and reverence, captured by the Syilx concept of Tmixw – the people only take the salmon that is needed

(Lower Nicola Indian Band (2015), ESSA Technologies Ltd. and Fraser Basin Council (2019), Band (2021)). The stewardship of their waters continues through the work of the Scw'exmx Tribal Council and its four member communities, the Lower Nicola Indian Band, and initiatives like the Nicola Watershed Governance Project. The Chinook Salmon (Endangered), Coho Salmon (Threatened), and steelhead (Endangered) populations have all been assessed and proposed for Species at Risk Act listing by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC). The development and implementation of this WCRP aims to support ongoing conservation and restoration work by the Nlaka'pamux/Scw'exmx, Syilx, and other local groups by providing an action plan to address the specific, tractable conservation threat posed by fragmentation, and conserving habitat connectivity for these important species.

Chinook Salmon | k' y'í e/pəqéłus | Oncorhynchus tshawytscha

Table 1.3: Chinook Salmon Designated Unit assessment in the
shed. Assessments were undertaken by the Committee
Wildlife in Canada (2020).

COSEWIC Designated	Status	Trend	Median percent change (last 3 gene
15 - Lower Thompson (Stream, Spring)	Endangered	TBD	TBD

Chinook Salmon are one of the first species to return to the watershed each year, arriving as early as May. The population has been in decline since the mid-1990s (LGL Ltd. (2007), Ecoscape Ltd. (2017)). Known and historic spawning locations include the mainstem Nicola River (mostly between the Coldwater River and Spius Creek confluences), Coldwater River, Spius Creek, lower portions of Clapperton Creek, and upstream of Nicola Lake in Moore Creek and the Upper Nicola River (LGL Ltd. (2007), Ecoscape Ltd. (2017)). In addition to these spawning locations, important juvenile rearing areas have been observed in Juliet Creek and Voght Creek (LGL Ltd. (2007)). Chinook Salmon stocks have been supplemented by the Spius Creek hatchery since the 1980s. Fry and smolt releases have both occurred, and the Nicola stock is currently enhanced by ~200,000 coded-wire-tagged yearling smolts released annually as a component of the Pacific Salmon Commission indicator stock study program (R. Bailey, pers. comm.).

Coho Salmon | sx̣ayqs | Oncorhynchus kisutch

Table 1.4: Coho Salmon Designated Unit assessment in the
Assessments undertaken by the Committee on the S
Canada (2016).

COSEWIC Designated Unit	Status	Trend	Median percent change (last 3
Interior Fraser – Lower Thompson population	Threatened	N/A	119%

Coho Salmon begin to return to the watershed in September, with spawning beginning in late October and continuing through December. The population has been in decline since the mid 1980s (DFO (1991), Ecoscape Ltd. (2017)). Coho Salmon stocks have been supplemented in recent decades by outplants from the Spius Creek Hatchery (Ecoscape Ltd. (2017)). The majority of Coho Salmon spawning and rearing occurs in the Coldwater River, Spius Creek, and Maka Creek, but has also been observed in Clapperton Creek, Prospect Creek, and the Upper Nicola River and Moore Creek (LGL Ltd. (2007), Pacific-Salmon-Foundation (2024)). The lower reaches of many smaller tributaries are also important Coho Salmon rearing habitat.

Steelhead | có łe | Oncorhynchus mykiss

Table 1.5: Steelhead Designated Unit assessment in the Lower Nico
sessments undertaken by the Committee on the Status o
Canada (2018).

COSEWIC Designated Unit	Status	Trend	Median percent change (last 3 generations)
Thompson River Population	Endangered	Declining	-79%

Steelhead join Chinook Salmon as the first to appear in the watershed, arriving in the spring (LGL Ltd. (2007)). The population is critically endangered and is seen as an extreme conservation concern (Fisheries and Oceans Canada (2018)). Steelhead spawning and rearing is known to occur in the lower portion of the mainstem Nicola River, Skuhun Creek, Shakan Creek, Nuaitch Creek, Maka Creek, Coldwater River, Juliet Creek, Voght Creek, Prospect Creek, and Clapperton Creek (LGL Ltd. (2007)). Historically, almost all third-order and greater streams would have supported steelhead, and groundwater-fed thermal refugia continue to provide important rearing and holding habitats (Ecoscape Ltd. (2017)).

Structure Types

Table 5 highlights structure types threatening anadromous salmonids in the watershed. The results of this assessment were used to inform the subsequent planning steps; and identify knowledge gaps where there are limited spatial data to inform the assessment for a specific structure type.

Table 1.6: Structure types in the Lower Nicola River watershed and barrier rating assessment results. For each structure type listed, 'Extent' refers to the proportion of key habitat for all focal species that is being blocked by that structure type, 'Severity' is the proportion of structures for each structure type that are known to block passage for focal species based on field assessments, and 'Irreversibility' is the degree to which the effects of a structure type can be reversed and connectivity restored. Key habitat for the purpose of this exercise is defined as the total amount of combined thermal refuge, spawning, and rearing habitat.

Structure Types	Extent	Severity	Irreversibility	Overall Threat Rating:
Lateral Barriers (including to thermal connectivity)	High	Very High	Medium	High
Natural Barriers	High	Very High	Medium	High
Road-Stream Crossings	High	Very High	Low	Medium
Small Dams(<5m height)	Low	Very High	Medium	Low
Abandoned Rail-stream Crossings (longitudinal)	Low	Low	Medium	Low
Trail-stream Crossings	Low	Low	Low	Low

2 Methods and Definitions

Connectivity Modelling

To prioritize culverts for remediation, connectivity status and structure ranks were produced using BCFishPass, an open-source freshwater connectivity modelling framework, which builds off provincial barrier mapping tools (Mount et al. 2011) and includes three sub-models:

Accessibility model: Naturally accessible waterbodies are those that are considered likely accessible to focal species if no human-made barriers existed on the landscape. These were spatially delineated for each focal species using natural barriers (i.e., waterfalls, gradient barriers, or subsurface flows) that would naturally limit upstream movement.

Habitat model: A subset of the naturally accessible waterbody layer was defined as key habitat, i.e., habitat likely to support spawning or rearing, rather than simply being used as a movement corridor. The habitat model identifies areas that have a higher potential to support key habitat based on stream channel gradient and discharge.

Connectivity model: A layer of known or modelled structures was overlaid on the key habitat model output. Modelled stream crossings were created from stream network and road/railway Geographic Information System (GIS) data, by mapping where a road or railway crosses a stream. Dam data were obtained from the Canadian Aquatic Barriers Database. All mapped dams and modeled crossings (henceforth structures) downstream of key habitat were initially considered potential barriers.

Some structures were classified as non-existent or passable prior to initiating field work and excluded from the model. Stream crossings on 6th order and larger streams were assumed to be bridges and classified as passable. Remaining structures were checked on Google Earth to verify they were not bridges, cross-ditches, or non-existent crossings. Data from previous field assessments was obtained from the Provincial Stream Crossing Inventory System (PSCIS), and structures that had been assessed and scored as passable were classified as such. Assessed structures that were scored as barriers or potential barriers were considered barriers in the model, along with all dams lacking fishways and structures that have not yet been assessed. Models were run to produce initial outputs, and maps were created showing disconnected key habitat and associated structures.

Local knowledge and data were incorporated into model development in two phases and overruled modeled outputs where the two differed. Existing reports on habitat, barriers, and fish

distributions, when available, were incorporated into the initial model outputs. Maps of initial outputs were then shared with local knowledge holders, who identified model errors or omissions (e.g., passable structures, or known spawning habitats that were not identified by the model). Corrections to model data were made based on local knowledge, and models were re-run.

Connectivity status was estimated by calculating the proportion of key habitat that is connected to the ocean (i.e., not fragmented by human-made barriers). Key habitat with no structures or only passable structures downstream was considered connected.

Ranking: Structures, and sets of structures that combine to disconnect habitat, were ranked by the amount of key habitat upstream. A ranked structure list helps determine where structures are blocking large amounts of habitat, and where removal of an individual structure or a group of structures (a "set") is more effective in opening up habitat, by ranking structures based on amount of habitat blocked. Top-ranked structures or sets of structures represented those with the most key habitat upstream and did not consider other factors such as habitat quality and cost, which ultimately drove restoration prioritization when planning field work and deciding priorities with partners.

Habitat Accumulation Curves (HACs): HACs are used to visualize structure rankings in order of the relative amount of habitat they are blocking upstream. Structures are plotted in order of km's blocked by structures. HACs illustrate restoration priorities by showing how much habitat becomes accessible as barriers are removed or improved. This method of visualizing cumulative habitat gains is useful for prioritizing restoration work and communicating results to partners.

Table 2.1

Focal Species	KEA	Indicator	Poor	Fair	Good	Very Good
Anadromous salmonids	Available off-channel Thermal Refuge				Total Area (m2) of thermal refuge accessible	
					Current Status:	

Comments: No baseline data exists on the extent of overwintering habitat in the watershed. A priority action is included in the Operational Plan (strategy 2.3) to develop a habitat layer, and this will be used to inform this connectivity status assessment in the future.

Table 2.2

Focal Species	KEA	Indicator	Poor	Fair	Good	Very Good
Anadromous salmonids	Available Spawning Habitat			% of total linear spawning habitat accessible		<25%
				Current Status:		

Comments: Indicator rating definitions are based on the consensus decisions of the planning team, including the decision not to define Fair. The current status is based on the connectivity model output, which is current as of December 2024.

Table 2.3

Focal Species	KEA	Indicator	Poor	Fair	Good	Very Good
Anadromous salmonids	Available Rearing Habitat	% of total linear rearing habitat accessible Current Status:	<25%	26-		

Comments: Indicator rating definitions are based on the consensus decisions of the planning team, including the decision not to define Fair. The current status is based on the connectivity model output, which is current as of December 2024.

Structure Ranking

Lateral Barriers (Including to thermal refugia)

There is a lack of comprehensive data and mapping of lateral barriers and potential thermal refugia in the watershed to currently support a strategic prioritization (see Table **??**). However, local knowledge was used to compile a list of candidate sites for field verification as a starting point to improve lateral and thermal connectivity. These sites were investigated in 2021 (Mozin (2021)), but an atmospheric river event later that winter resulted in major modifications and rearrangement of the Coldwater River and mainstem Nicola River downstream of the Coldwater confluence, rendering the outcome of those investigations no longer relevant. Further assessment of off-channel thermal refugia was undertaken by Nooaitch Band in 2023, with results of those assessments expected to be released to the broader community working on conservation in the Lower Nicola Watershed in the near future.

Longitudinal Field Assessment Ranking Process

There is a lack of comprehensive data and mapping of lateral barriers and potential thermal refugia in the watershed to currently support a strategic prioritization (see Table **??**). However, local knowledge was used to compile a list of candidate sites for field verification as a starting point to improve lateral and thermal connectivity. These sites were investigated in 2021 (Mozin (2021)), but an atmospheric river event later that winter resulted in major modifications and rearrangement of the Coldwater River and mainstem Nicola River downstream of the Coldwater confluence, rendering the outcome of those investigations no longer relevant. Further assessment of off-channel thermal refugia was undertaken by Nooaitch Band in 2023,

with results of those assessments expected to be released to the broader community working on conservation in the Lower Nicola Watershed in the near future.

Table 2.4: Rearing habitat connectivity gain requirements to meet WCRP goals in the Lower Nicola River watershed. The measures of currently accessible and total habitat values are derived from the intrinsic potential habitat model

Habitat Type	Currently connected (km)	Total (km)	Current Connectivity Status	Goal	Gain required
Spawning and Rearing	411.42	455.59	90%	90%	25.95

Longitudinal Structure Prioritization Summary

There is a lack of comprehensive data and mapping of lateral barriers and potential thermal refugia in the watershed to currently support a strategic prioritization (see Table **??**). However, local knowledge was used to compile a list of candidate sites for field verification as a starting point to improve lateral and thermal connectivity. These sites were investigated in 2021 (Mozin (2021)), but an atmospheric river event later that winter resulted in major modifications and rearrangement of the Coldwater River and mainstem Nicola River downstream of the Coldwater confluence, rendering the outcome of those investigations no longer relevant. Further assessment of off-channel thermal refugia was undertaken by Nooaitch Band in 2023, with results of those assessments expected to be released to the broader community working on conservation in the Lower Nicola Watershed in the near future.

Barrier Id	Modelled Crossing Id	Watercourse Name	Road Name	Structure Type
197,882		Stumbles Creek	Unnamed	Stream crossing - CBS
197,884	11,300,751	Stumbles creek	Highway 8	Stream crossing - CBS

Barrier Id	Modelled Crossing Id	Watercourse Name	Roa
196942	11,300,383	Mine Creek	Mine
203627	11,300,057	Mine Creek	High
203641	24,758,913	Stumbles Creek	Priva
203642		Stumbles Creek	Priva

Barrier Id	Modelled Crossing Id	Watercourse Name	Roa
203637		Stumbles Creek	Grav
203639		Stumbles Creek	Unn
203634	11,303,636	Shackan Creek	Unn
196966	11,300,830	Bottletop Creek	Julie
203630	11,300,395	Bottletop Creek	High
197681	11,304,582	Chataway Creek	Skul
197692	11,305,296	Manning Creek	Man
197693	11,303,684	Manning Creek	Unn
203629	11,302,252	Trib to Coldwater	Kettl
6ac0dab8-0785-4e46-ac24-c99d538b05bf		Chataway Creek	n/a
1011302252		Trib to Coldwater @ KVR	

Barrier Id	Modelled Crossing Id	Watercourse Name	Road Name
1011304238	11,304,238	Skuhun Creek	Mouth
197889	24,713,603	Skuhun Creek	Highway 8
36b0bcda-4671-45ff-8dea-d1fc8abe3a8d		Clapperton Creek	n/a
196969	11,304,548	Murray Lake Creek	Maka-Murra

Glossary

Freshwater connectivity

The degree to which aquatic species can disperse and/or migrate freely through freshwater systems.

Longitudinal connectivity

Connectivity of a stream along the upstream-downstream plane, including access to tributaries and spawning and rearing habitat. Longitudinal connectivity can be fragmented by physical barriers (e.g., anthropogenic, or natural structures) or by physiological limits of distribution for species (e.g., stream gradient, temperature, or flow requirements).

Lateral connectivity

Connectivity of a stream bed to adjacent riparian wetlands, floodplains, side-channels, or off-channel features, including access to rearing and overwintering habitat. Lateral connectivity can be fragmented by physical barriers, channelization, armoring of the stream bed, or artificial flow regulation.

Vertical connectivity

Connectivity of a stream bed to groundwater/hyporheic zone, including access to oxygen-rich water and temperature refugia. Vertical connectivity can be fragmented by water withdrawals and anthropogenically induced changes to the hydrological, thermal, and sediment regimes of the watershed.

Temporal connectivity

Connectivity variability in any of the three spatial dimensions (longitudinal, lateral, vertical) based on changes in the natural flow regime over time. Variation in the temporal connectivity occurs naturally; however, fragmentation can be exacerbated through anthropogenically induced changes to the hydrological, thermal, and sediment regimes of the watershed.

Geographic Scope

The spatial bounds within which connectivity models are applied and fish-passage improvement efforts are focused. This area is refined after the accessibility model is developed to cover only streams that are naturally accessible to each focal species or guild (see naturally accessible streams below).

Focal Species

The ecologically and culturally important species (e.g., Chinook Salmon) for which habitat connectivity is being conserved and/or restored in the watershed.

Focal Guild

Two or more focal species grouped together by shared characteristics (e.g., migratory behavior, swimming ability, and habitat use).

Key Ecological Attribute (KEA)

An aspect of a focal species or guild's biology or ecology that, if missing or altered, would lead to the loss of that focal species over time (e.g., proportion of connected spawning habitat).

Structure

A feature, either anthropogenic (e.g., culvert, dam) or a temporary natural feature (e.g., barrier beach, debris jam) that may fragment freshwater systems and limit the ability of species to disperse and/or migrate freely within their naturally accessible habitat. Differentiated from more permanent natural features, such as waterfalls, that delineate naturally accessible waterbodies.

Set of structures

A set of two or more structures that must be addressed/considered together for restoration or rehabilitation when considering the total amount of connectivity to be gained during barrier prioritization.

Current Connectivity Status

The measure of important habitat (i.e., amount, proportion, or index) that is currently connected (i.e., unaffected by structures with a status of unknown, partially passable, or impassable) for each focal species. For non-diadromous species, indices such as the longest fragment or Dendridic Connectivity Index – potadromous (DCIp; Cote et al. 2009).

Connectivity Goal

A SMART (Specific, Measurable, Achievable, Relevant, and Time-Bound) goal representing the desired future connectivity status of each focal species.

Structure Prioritization

An analysis that ranks structures (individually and/or in sets) by the amount of key habitat disconnected by each individual structure. Structures are ranked in descending order according connectivity gains (e.g., greatest amount of linear km of important habitat) if restored or rehabilitated. This analysis includes structures with a passability status of "unknown", "partially passable", or "impassable".

Restoration

Returning the ecological structure and function (e.g., natural channel-forming processes) of a site to pre-disturbance conditions.

Rehabilitation

The partial recovery of ecosystem function. This is typical when improving passage by replacing one type of crossing with another that is passable but may still affect the streambed or channel-forming processes.

Rehabilitated barriers

Structures for which connectivity for aquatic species was restored by altering or maintaining a structure that previously impeded movement.

Remediation

A term for a form of restoration most often associated with mining, referring to the remediation of contaminated sediments. Often used as jargon in fish-passage discussions because it applies to both fish passage and barriers (i.e., both can be described as "remediated"). Most fish-passage projects involve ecological rehabilitation, but it is inaccurate to write that a barrier was rehabilitated. Instead, fish passage should be described as rehabilitated, improved, or restored (if appropriate), and barriers described as removed or addressed.

Excluded structures

Structures that were investigated during a project but were removed from consideration or the model. Reasons for their exclusion could be that they do not exist, are passable, or no key habitat was found upstream.

Assessed structures that remain data deficient

Structures that have had some form of assessment but need further investigation before a decision can be made to categorize them as either confirmed barriers, excluded structures (passable, non-existent, no suitable upstream habitat), or rehabilitated structures.

3 Connectivity Status and Ranking

Current Connectivity Status

Table 3.1

Focal Species	KEA	Indicator	Poor	Fair	Good	Very Good
Anadromous salmonids	Available off-channel Thermal Refuge				Total Area (m2) of thermal refuge accessible Current Status:	

Comments: No baseline data exists on the extent of overwintering habitat in the watershed. A priority action is included in the Operational Plan (strategy 2.3) to develop a habitat layer, and this will be used to inform this connectivity status assessment in the future.

Table 3.2

Focal Species	KEA	Indicator	Poor	Fair	Good	Very Good
Anadromous salmonids	Available Spawning Habitat			% of total linear spawning habitat accessible Current Status:		<25%

Comments: Indicator rating definitions are based on the consensus decisions of the planning team, including the decision not to define Fair. The current status is based on the connectivity model output, which is current as of December 2024.

Table 3.3

Focal Species	KEA	Indicator	Poor	Fair	Good	Very Good
Anadromous salmonids	Available Rearing Habitat			% of total linear rearing habitat accessible Current Status:	<25%	26-

Comments: Indicator rating definitions are based on the consensus decisions of the planning team, including the decision not to define Fair. The current status is based on the connectivity model output, which is current as of December 2024.

Table 3.4: Goals to improve (1) off-channel thermal refuge, (2) spawning, and (3) rearing habitat connectivity for focal species in the Lower Nicola River watershed over the lifespan of the WCRP (2021-2031). The goals were established through discussions with the planning team and represent the resulting desired state of connectivity in the watershed. The goals are subject to change as more information and data are collected over the course of the plan timeline (e.g., the current connectivity status is updated based on barrier field assessments).

Goal #	Goal
1	By 2031, the total area of groundwater-serviced off-channel thermal refuge accessible to anadromous
2	By 2025, the % of total linear spawning habitat accessible to anadromous salmonids will not decreas
3	By 2031, the % of total linear rearing habitat accessible to anadromous salmonids will increase from

Ranked Structures Table

Table 3.5: Ranked Structures List

Barrier ID	Site Name	Structure Status	Upstream Habitat Rank	Avg Habitat Gain	Priority	Next Ste
36b0bcda-4671-45ff-8dea-d1fc8abe3a8d	Clapperton Dam	Rehabilitated barrier				
197889	Skuhun Creek @ Hwy 8	Rehabilitated barrier				
1011304238	Skuhun Creek @ Mouth	Rehabilitated barrier				
196969	Murray Lake Creek	Rehabilitated barrier				
197882	Stumbles Creek @ private (1011304215)	Confirmed barrier				
197884	Stumbles @ Hwy 8 (1011300751)	Confirmed barrier				
196942	Mine Creek @ Mine Creek Road	Assessed structure that re				
203627	Mine Creek at Highway 5	Assessed structure that re				
197041	Godey Creek @ Pooley Road	Excluded structure				
203634	Shackan Creek @ unnamed rd	Assessed structure that re				
1024731185	None	Excluded structure				
1011300553	None	Excluded structure				
1011306461	Salem Creek @ KVR	Excluded structure				
203637	Stumbles Creek @ Gravel Pit	Assessed structure that re				
203639	Stumbles Creek @ Unnamed	Assessed structure that re				
203641	Stumbles Creek @ Private	Assessed structure that re				
203642	Stumbles Creek @ Private	Assessed structure that re				
196966	Bottletop Creek @ Juliet Dr	Assessed structure that re				
203630	Bottletop Creek @ Hwy 5	Assessed structure that re				
1011300018	None	Excluded structure				
1011300705	None	Excluded structure				
1011300844	None	Excluded structure				

Barrier ID	Site Name	Structure Status	Upstream Habitat Rank	Avg Habitat Gain	Priority	Next Ste
1011302471			None		Excluded structure	
196957			None		Excluded structure	
196997			None		Excluded structure	
197012			None		Excluded structure	
197014			None		Excluded structure	
197036			None		Excluded structure	
197039			None		Excluded structure	
197040			None		Excluded structure	
197042			Godey Creek at Sunnyview Street		Excluded structure	
197043			Godey Creek at Coldwater Road		Excluded structure	
197044			Godey Creek at Sterling Road		Excluded structure	
197681			Chataway @ Skuhun-Pimanus FSR		Assessed structure that re	
197692			Manning Creek @ Manning FSR		Assessed structure that re	
197693			Manning Creek @ Private		Assessed structure that re	
197694			None		Excluded structure	
197695			None		Excluded structure	
197696			None		Excluded structure	
197880			None		Excluded structure	
197881			None		Excluded structure	
197883			None		Excluded structure	
197885			None		Excluded structure	
197886			None		Excluded structure	
197887			None		Excluded structure	
197888			None		Excluded structure	
197890			None		Excluded structure	
197891			None		Excluded structure	
198881			None		Excluded structure	
198882			1011303627		Excluded structure	
198883			1011303791		Excluded structure	
203621			Trib 1 to Coldwater @ Pipeline		Excluded structure	
203624			Trib 1 to Coldwater @ Hwy 5		Excluded structure	
203625			11301334		Excluded structure	
203628			Mine Creek at KVR		Excluded structure	
203629			Trib 1 to Coldwater		Assessed structure that re	
203633			11304154		Excluded structure	
203636					Excluded structure	
203638			11302218		Excluded structure	
203640			1011300123		Excluded structure	
55cd5d66-5bef-446f-bce3-8b097d001454			1100002544		Excluded structure	
6ac0dab8-0785-4e46-ac24-c99d538b05bf			Chataway weir		Assessed structure that re	
d0fe7d87-bba1-4edb-b132-01bdeccd7277			1100002545		Excluded structure	